

BREAK-EVEN ANALYSIS

Study Guide • Worked Examples • Practice

Learning Goal

By the end of this handout, you should be able to identify fixed and variable costs, calculate total cost and revenue, find the break-even quantity, calculate profit or loss, and compare business alternatives.

1. What Is the Break-Even Point?

The break-even point (BEP) is the level of sales where a business makes exactly enough revenue to cover all of its costs. At this point, profit is zero: the business is not making a profit, but it is also not making a loss.

Think of it this way

Below the break-even point → LOSS

At the break-even point → PROFIT = 0

Above the break-even point → PROFIT

2. Key Terms You Must Know

Term	Meaning	Example
Fixed Cost (Fc)	Cost that does not change with production volume.	Rent, equipment, salaries
Variable Cost per Unit (Cv)	Cost added for each unit produced or sold.	\$65 per table
Selling Price per Unit (P)	Amount received from selling one unit.	\$180 per table
Volume (V)	Number of units produced and sold.	300 tables
Total Variable Cost	Variable cost per unit × quantity.	65×300
Total Cost (TC)	Fixed cost + total variable cost.	$F + Qv$
Total Revenue (TR)	Selling price × quantity sold.	Qp
Contribution Margin	Amount each unit contributes toward fixed cost and then profit.	$p - v$
Profit (Z)	Revenue remaining after all costs.	$TR - TC$

3. Core Formulas

Total Revenue: $TR = V \times p$

Quantity sold multiplied by selling price per unit.

Total Cost: $TC = Fc + (V \times Cv)$

Fixed cost plus total variable cost.

Profit: $Profit = TR - TC = V.p - F - V.Cv$

Contribution Margin per Unit: $CM = p - v$

Break-Even Quantity: $V = F \div (p - Cv)$

Important condition

For a normal break-even quantity to exist, the selling price must be greater than the variable cost per unit ($p > v$). If $p \leq v$, each unit does not contribute enough to cover fixed costs.

4. A Simple 5-Step Method

Step 1 - List the data: Write F , v , p , and Q clearly before calculating.

Step 2 - Find contribution margin: Calculate $p - v$.

Step 3 - Choose the correct formula: Use TR , TC , profit, or break-even depending on the question.

Step 4 - Calculate carefully: Keep units consistent and show your substitutions.

Step 5 - Interpret the result: Explain what the number means in business terms, not only mathematically.

5. Worked Example 1 - Willow Furniture Company

The Willow Furniture Company produces tables. Fixed monthly production cost is \$8,000, variable cost is \$65 per table, and each table sells for \$180. For a monthly volume of 300 tables:

Given

$F_c = \$8,000$ $C_v = \$65/\text{table}$ $P = \$180/\text{table}$ $V = 300 \text{ tables}$

A. Total Cost, Total Revenue, and Profit

Step 1 - Total variable cost: $V \times C_v = 300 \times \$65 = \$19,500$

Step 2 - Total cost: $TC = F + V.C_v = \$8,000 + \$19,500 = \$27,500$

Step 3 - Total revenue: $TR = V.p = 300 \times \$180 = \$54,000$

Step 4 - Profit: $\text{Profit} = TR - TC = \$54,000 - \$27,500 = \$26,500$

Answer / Interpretation: At 300 tables per month, Willow Furniture earns a monthly profit of \$26,500.

B. Break-Even Volume

Step 1 - Contribution margin: $p - v = \$180 - \$65 = \$115 \text{ per table}$

Step 2 - Break-even quantity: $V = \$8,000 \div \$115 = 69.57 \text{ tables}$

Step 3 - Round up: The company cannot normally sell 0.57 of a table, so round up to 70 tables.

Answer / Interpretation: Willow Furniture must sell at least 70 tables per month to cover all costs.

Beginner Note

Do not round 69.57 down to 69. Rounding down would leave the company slightly below break-even.

6. Worked Example 2 - Pastureland Dairy

Pastureland Dairy has fixed monthly cost of \$4,000 and variable cost of \$0.21 per pound. At the current price of \$0.75, it sells 9,000 pounds. It is considering a price of \$0.95, but expected sales would fall to 5,700 pounds.

Decision Question

Should the dairy raise the price? Compare PROFIT under both alternatives.

Case 1: Current Price

$$\text{Profit} = V.p - F - V.Cv = 9,000(\$0.75) - \$4,000 - 9,000(\$0.21)$$

$$\text{Profit} = \$6,750 - \$4,000 - \$1,890 = \$860$$

Case 2: Higher Price

$$\text{Profit} = 5,700(\$0.95) - \$4,000 - 5,700(\$0.21)$$

$$\text{Profit} = \$5,415 - \$4,000 - \$1,197 = \$218$$

Answer / Interpretation: Based on these estimates, the dairy should NOT raise the price: estimated monthly profit falls from \$860 to \$218, a decrease of \$642.

Decision-Making Lesson

A higher selling price does not automatically mean higher profit. The change in sales volume must also be considered.

7. Worked Example 3 - Online MBA Program

The College of Business is considering an online MBA program. Start-up cost is \$350,000. Tuition is \$18,000 per student per year, while the university charges the college \$12,000 per enrolled student.

Given

$$F = \$350,000 \quad p = \$18,000/\text{student} \quad v = \$12,000/\text{student}$$

A. Students Needed to Break Even

$$\text{Contribution margin} = \$18,000 - \$12,000 = \$6,000 \text{ per student}$$

$$V = \$350,000 \div \$6,000 = 58.33 \text{ students}$$

Answer / Interpretation: The college needs at least 59 students to break even. Because students are whole people, 58.33 must be rounded up.

B. Profit with 75 Students

$$\text{Profit} = 75(\$18,000) - \$350,000 - 75(\$12,000)$$

$$\text{Profit} = \$1,350,000 - \$350,000 - \$900,000 = \$100,000$$

Answer / Interpretation: With 75 students, the program earns \$100,000.

C. Tuition Increased to \$24,000, Enrollment Falls to 35

$$\text{Profit} = 35(\$24,000) - \$350,000 - 35(\$12,000)$$

$$\text{Profit} = \$840,000 - \$350,000 - \$420,000 = \$70,000$$

Answer / Interpretation: The proposed higher tuition gives an estimated profit of \$70,000, compared with \$100,000 at 75 students and \$18,000 tuition. Under these assumptions, the original plan produces \$30,000 more profit.

8. Extra Easy Example - Coffee Cart

A coffee cart pays \$1,200 per month in fixed costs. Each cup costs \$1.50 to make and sells for \$4.50.

$$\text{Contribution margin} = \$4.50 - \$1.50 = \$3.00 \text{ per cup}$$

$$\text{Break-even quantity} = \$1,200 \div \$3.00 = 400 \text{ cups}$$

Answer / Interpretation: The coffee cart breaks even at 400 cups per month. Cup 401 begins to contribute to profit.

9. Extra Example - Target Profit

Suppose the coffee cart wants a monthly profit of \$900. How many cups must it sell?

$$\text{Units for Target Profit: } Q = (F + \text{Target Profit}) \div (p - v)$$

$$Q = (\$1,200 + \$900) \div \$3.00 = 700 \text{ cups}$$

Answer / Interpretation: The cart must sell 700 cups to earn a \$900 monthly profit.

10. Common Mistakes to Avoid

Mistake	How to Fix It
Mixing total variable cost with variable cost per unit	Use v for cost per unit; Qv is total variable cost.
Using $p + v$ in the break-even formula	The denominator is contribution margin: $p - v$.
Forgetting fixed cost	Profit must subtract both fixed cost and total variable cost.
Rounding break-even down	For whole units, round UP so all costs are covered.
Comparing revenue instead of profit	A decision can increase revenue but still reduce profit.
Giving only a number	Always interpret: 'The company must sell at least ____ units.'

11. Quick Formula Cheat Sheet

What You Need	Formula
Total Revenue	$TR = Qp$
Total Variable Cost	$TVC = Qv$
Total Cost	$TC = F + Qv$
Profit	$\text{Profit} = Qp - F - Qv$
Contribution Margin / unit	$CM = p - v$
Break-Even Quantity	$Q^* = F \div (p - v)$
Units for Target Profit	$Q = (F + \text{Target Profit}) \div (p - v)$

12. Practice Problems

1. A T-shirt business has fixed costs of \$2,400 per month. Each shirt costs \$8 to produce and sells for \$20. Find the break-even quantity.
2. A bakery has fixed costs of \$3,000, variable cost of \$2 per cake, and selling price of \$8 per cake. If it sells 700 cakes, find total revenue, total cost, and profit.
3. A small manufacturer has fixed costs of \$15,000, selling price of \$50 per unit, and variable cost of \$30 per unit. How many units are needed to earn a target profit of \$5,000?
4. A product currently sells for \$25 with variable cost \$10 and monthly fixed cost \$6,000. At 600 units, calculate profit. Then explain whether the business is above or below break-even.

13. Self-Check Quiz

1. At the break-even point, profit equals: (a) fixed cost (b) zero (c) revenue
2. Which cost stays constant in the basic model? (a) fixed cost (b) total variable cost (c) revenue
3. Contribution margin per unit equals: (a) $p + v$ (b) $F \div Q$ (c) $p - v$
4. If selling price increases while all other inputs stay unchanged, what normally happens to break-even quantity?
5. True or False: A higher selling price always produces a higher total profit.

14. Answer Key

Practice 1: $CM = \$20 - \$8 = \$12$; $BEP = \$2,400 \div \$12 = 200$ shirts.

Practice 2: $TR = 700 \times \$8 = \$5,600$; $TC = \$3,000 + 700 \times \$2 = \$4,400$; Profit = \$1,200.

Practice 3: $Q = (\$15,000 + \$5,000) \div (\$50 - \$30) = 1,000$ units.

Practice 4: Profit = $600(\$25) - \$6,000 - 600(\$10) = \$3,000$. $BEP = \$6,000 \div \$15 = 400$ units, so 600 units is above break-even.

Quiz: 1-b; 2-a; 3-c; 4-Break-even quantity normally decreases; 5-False.

Final Study Tip

When you see a break-even question, first write F, v, p, and Q. Then calculate contribution margin ($p - v$). That one step makes most break-even problems much easier.

15. Applied Example - Break-Even Analysis with Python

Python can perform the same break-even calculations that we solved by hand. This is useful when a business wants to test many different prices, costs, or sales volumes quickly.

We will reuse the Willow Furniture example

Fixed cost = \$8,000

Variable cost = \$65 per table

Selling price = \$180 per table

Sales volume = 300 tables

A. Basic Python Calculation

The program below calculates total variable cost, total cost, total revenue, profit, contribution margin, and the break-even number of tables.

```
# Break-Even Analysis: Willow Furniture
import math

# Step 1: Enter the known information
fixed_cost = 8000
variable_cost = 65
selling_price = 180
quantity = 300

# Step 2: Calculate total variable cost
total_variable_cost = quantity * variable_cost

# Step 3: Calculate total cost and total revenue
total_cost = fixed_cost + total_variable_cost
total_revenue = quantity * selling_price

# Step 4: Calculate profit
profit = total_revenue - total_cost

# Step 5: Calculate contribution margin per table
contribution_margin = selling_price - variable_cost

# Step 6: Calculate break-even quantity
break_even_exact = fixed_cost / contribution_margin
break_even_units = math.ceil(break_even_exact)

# Step 7: Display the results
print("Total Variable Cost = $", total_variable_cost)
print("Total Cost = $", total_cost)
print("Total Revenue = $", total_revenue)
print("Profit = $", profit)
print("Contribution Margin = $", contribution_margin)
print("Exact Break-Even Quantity =", round(break_even_exact, 2))
print("Minimum Whole Tables to Break Even =", break_even_units)
```

Expected Output

```
Total Variable Cost = $ 19500
Total Cost = $ 27500
Total Revenue = $ 54000
```

Profit = \$ 26500
Contribution Margin = \$ 115
Exact Break-Even Quantity = 69.57
Minimum Whole Tables to Break Even = 70